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Essentials of Data Science Laboratory - 2304102L - 2026 - 2304102L

Dashboard Contents Calendar Assessments Syllabus

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Course progress
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Resume ›

Description

Essentials of Data Science Laboratory - 2304102L

Upcoming tests

No upcoming exams in the next 7 days

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1.1.1. Calculate Momentum

04:04

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

m is the mass of the object (in kilograms).

v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Explorer

calculate...

Submit

```
#Type Content here...
m=float(input())
v=float(input())
M=m*v
```

Debugger

Average time	Maximum time
0.021 s	0.046 s
21.50 ms	46.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 46 ms Debug

Expected output	Actual output
5.0	5.0
10.0	10.0
50.00kgm/s	50.00kgm/s

Test case 2 17 ms

Terminal Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits 19:32

Write a Python program that accepts an integer n as input. Depending on the number of digits in n.

Constraints:
$1 \le n \le 999$

Input Format:
- The input consists of a single integer n.

Output Format:
- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases +

Explorer condition... Submit Debugger

```python
#write your code here...
import math
n=int(input())
if(n>=1 and n<=9):
```

Average time	Maximum time
0.008 s	0.009 s
7.57 ms	9.00 ms

4 out of 4 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 9 ms Debug

Expected output	Actual output
9	9
81	81

Test case 2 9 ms

Test case 3 6 ms

Terminal Test cases

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1.1.3. Days between Two Dates

56:22

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY - MM - DD$.
- The second line contains the second date in the format $YYYY - MM - DD$.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases +

noOfDay... Submit

```
from datetime import datetime
date1=input().strip()
date2=input().strip()
d1=datetime.strptime(date1, "%Y-%m-%d")
```

Average time 0.030 s 29.75 ms

Maximum time 0.080 s 80.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 80 ms Debug

Expected output	Actual output
2014-07-02	2014-07-02
2014-11-02	2014-11-02
123	123

Test case 2 11 ms

Terminal Test cases

1.1.4. Reverse a Number

04:09

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases +

reverseN... Submit

```
#write your code here...
n=int(input())
str=(str(n)[::-1])
rev=int(str)
```

Average time 0.011 s 11.20 ms

Maximum time 0.017 s 17.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 17 ms Debug

Expected output	Actual output
5367	5367
7635	7635

Test case 2 7 ms

Terminal Test cases

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1.1.5. Multiplication Table 02:56

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases +

Explorer multiplica... Submit Debugger

```python
#write your code here...
n=int(input())
for i in range (1,11,1):
    print(f"{n} x {i} = {n*i}")
```

Average time	Maximum time
0.007 s	0.007 s
6.75 ms	7.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 7 ms Debug

Expected output	Actual output
8	8
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16
8 x 3 = 24	8 x 3 = 24
8 x 4 = 32	8 x 4 = 32
8 x 5 = 40	8 x 5 = 40

Terminal Test cases

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1.2.1. Pass or Fail 05:56

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n, the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).

Explorer passorFa... Submit Debugger

```
# write your code here
n=int(input())
marks=list((map(int,input().split())))
if any(mark<40 for mark in marks):
```

Average time 0.013 s 12.60 ms

Maximum time 0.017 s 17.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 17 ms Debug

Expected output	Actual output
4	4
55 65 78 89	55 65 78 89
Aggregate Percentage: 71.75	Aggregate Percentage: 71.75
Grade: First Division	Grade: First Division

Test case 2 12 ms

Terminal Test cases

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1.2.2. Fibonacci Series 02:05

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases +

Explorer fibonacci.... Submit Debugger

```
def fibonacci(n):
    if n==1:
        return 0
    elif n==2:
```

Average time	Maximum time
0.018 s	0.049 s
18.50 ms	49.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 9 ms Debug

Expected output	Actual output
5	5
0 1 1 2 3	0 1 1 2 3

Test case 2 7 ms

Terminal Test cases

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1.2.2. Fibonacci Series 02:05

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Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases +

Explorer fibonacci.... Submit Debugger

```
def fibonacci(n):
    if n==1:
        return 0
    elif n==2:
```

Average time	Maximum time
0.018 s	0.049 s
18.50 ms	49.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 9 ms Debug

Expected output	Actual output
5	5
0 1 1 2 3	0 1 1 2 3

Test case 2 7 ms

Terminal Test cases

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1.2.3. Pattern - 1

01:46

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

rightangl...

Submit

```
#Type Content here...
n=int(input())
for i in range(1,n+1):
    for j in range(i):
```

Average time	Maximum time
0.007 s	0.009 s
7.17 ms	9.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 9 ms Debug

Expected output	Actual output
5	5
*	*
* *	* *
* * *	* * *
* * * *	* * * *
* * * * *	* * * * *

Terminal Test cases

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1.2.4. Pattern - 2 02:40

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

numberP... Submit

```
# Type Content here...
n=int(input())
for i in range(1,n+1):
    for j in range(1,i+1):
```

Average time	Maximum time
0.008 s	0.009 s
7.75 ms	9.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 8 ms Debug

Expected output	Actual output
5	5
1	1
1 2	1 2
1 2 3	1 2 3
1 2 3 4	1 2 3 4
1 2 3 4 5	1 2 3 4 5

Terminal Test cases

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